
Legal RAG System

AI-Powered Due Diligence for Pakistani Law Firms

Day 1 Submission — Problem Design & Solution Architecture

Program	AI/ML Internship Program
Team	Pair B
Domain	Artificial Intelligence / Machine Learning
Sprint	5-Day Build
Target Industry	Corporate & Real Estate Law Firms
Markets	Islamabad, Rawalpindi, Lahore, Karachi
Date	June 30, 2026
Status	Day 1 Complete

5–7 Days	PKR 250K/week	48 Hours
Current Turnaround	Senior Associate Cost on Review	Client Expectation

Contents

1	Problem Statement	3
1.1	The Challenge	3
1.2	Why This Problem Is Specific to Pakistan	3
1.3	Who Is Affected	4
1.4	Cost of Inaction — Research Findings	4
1.4.1	Direct Revenue Loss	4
1.4.2	Wasted Senior Associate Time	4
1.4.3	Professional Liability Exposure	4
1.4.4	Competitive Displacement	4
1.4.5	Client Attrition Cost	5
2	Proposed Solution & Technical Architecture	5
2.1	Solution Overview	5
2.2	System Architecture	6
2.3	Dual-Index Architecture — Core Differentiator	6
2.4	Pakistan-Specific Intelligence Layer	7
2.4.1	Layer 1 — User Interface (React + Vite)	8
2.4.2	Layer 2 — Backend Orchestration (FastAPI)	8
2.4.3	Layer 3 — Document Processing (pdfplumber + pytesseract)	8
2.4.4	Layer 4 — Text Chunking (LlamaIndex SentenceSplitter)	8
2.4.5	Layer 5 — Embedding Generation (HuggingFace)	9
2.4.6	Layer 6 — Vector Storage (ChromaDB)	9
2.4.7	Layer 7 — Query Engine (LlamaIndex RouterQueryEngine)	9
2.4.8	Layer 8 — LLM Generation (Claude Sonnet 4.6 / Groq)	9
2.4.9	Layer 9 — Output Generation (python-docx + ReportLab)	9
2.5	Technology Stack & Selection Rationale	10
2.5.1	LlamaIndex (over LangChain)	10
2.5.2	Claude Sonnet 4.6 (Production LLM)	11
2.5.3	Groq Llama 3.3 70B (Development & Testing)	11
2.5.4	HuggingFace paraphrase-multilingual-MiniLM-L12-v2 (Embeddings)	11
2.5.5	ChromaDB (Vector Store)	11
2.5.6	FastAPI (Backend)	11
3	Due Diligence Checklist — Property Transaction	12
3.1	Property Transaction Checklist	12
3.2	Red Flag Detection Patterns	12
3.3	LLM Response Schema	13
4	Cost Analysis	14
4.1	Per-Review API Cost Model	14
4.2	Monthly SaaS Cost Projections	15
4.3	Pricing Model	15
4.3.1	Client ROI at Professional Tier	15

5	Target Client Identification	15
5.1	Primary Targets for Day 4 Demo	15
5.1.1	Target 1 — Rana Ijaz & Partners (Islamabad)	15
5.1.2	Target 2 — M.A. Awan & Co. / MAACO (Rawalpindi & Islamabad)	16
5.2	Backup Target	16
5.3	Approved Outreach Script	16
5.4	Client Feedback Form (Post-Demo)	17

1 Problem Statement

1.1 The Challenge

Mid-size law firms across Pakistan—particularly those operating in corporate law, real estate transactions, and financial advisory—face a significant operational bottleneck in their document review and due diligence processes. When a high-value transaction arrives (a property acquisition, a merger, or a loan agreement), junior associates are tasked with manually reviewing large bundles of legal documents, which may run into hundreds of pages, in order to identify specific clauses, flag inconsistencies, locate missing information, and produce a structured review memorandum for senior partners.

This process is inherently slow, resource-intensive, and prone to human error. Junior lawyers are often inconsistent in what they flag, frequently miss subtle contradictions across documents, and produce review memos of uneven quality. Senior partners must then spend considerable time correcting and augmenting this work—time that carries a high opportunity cost given their billable rates.

Core Pain Point

A typical due diligence review involving 30–50 pages of legal documents currently takes a junior associate team **5 to 7 working days** to complete. Clients, however, demand turnaround within **48 hours** for competitive transactions. This gap in delivery speed is directly causing client dissatisfaction and, in some cases, **loss of mandates** to larger firms with greater associate bandwidth.

1.2 Why This Problem Is Specific to Pakistan

The legal technology landscape in Pakistan remains nascent. Unlike law firms in the United Kingdom or the United States—where AI-assisted due diligence tools from vendors such as Kira Systems and Luminance are now standard—Pakistani law firms have virtually no access to locally configured, affordable, and contextually appropriate legal AI tooling.

Key Pakistan-specific factors that make off-the-shelf international solutions unsuitable:

- **Mixed-language documents:** Urdu-language clauses are frequently embedded within otherwise English documents, confusing international NLP models.
- **Local statute references:** Pakistani law references the CPC, PPC, Transfer of Property Act 1882, LDA/CDA/TMA regulations, and SECP filings that international models do not recognise.
- **Document format variance:** Local property documents (Fard, Intiqal, Registry) follow formats absent from international training data.
- **Cost barrier:** Enterprise legal AI platforms cost PKR 500,000+ per month in licensing fees—prohibitive for firms of 5–30 lawyers. This creates a clear market gap that a purpose-built, locally configured solution can fill at a fraction of the cost.

1.3 Who Is Affected

Primary Stakeholders:

- Managing Partners of 5–30 lawyer corporate or real estate law firms in Islamabad, Rawalpindi, Lahore, and Karachi
- Legal Heads at banks and financial institutions responsible for reviewing loan and security documentation
- Senior Associates who spend the majority of their time on first-pass document review rather than higher-value legal reasoning
- Clients (businesses, developers, financial institutions) who lose competitive advantage due to slow due diligence timelines

1.4 Cost of Inaction — Research Findings

Why Firms Cannot Afford to Wait

The cost of inaction is not hypothetical. It compounds monthly across four distinct dimensions: direct revenue loss, wasted labour, professional liability exposure, and long-term competitive displacement.

1.4.1 Direct Revenue Loss

A mid-size law firm handling 8–12 transactions per month, each worth PKR 500,000–1,500,000 in fees, that loses even 2 mandates per month to faster competitors suffers a revenue loss of **PKR 1M–3M per month** (PKR 12M–36M annually). This is the primary commercial risk of inaction.

1.4.2 Wasted Senior Associate Time

PKR 250,000 per week in senior associate time is spent on first-pass review work that this system automates. At 52 weeks annually, that is **PKR 13,000,000 per year per firm** in high-cost labour performing tasks automatable at PKR 2,000–5,000 per month in API costs—a cost ratio exceeding 200:1.

1.4.3 Professional Liability Exposure

A missed clause—an undisclosed encumbrance, a defective title chain, a missing NOC—in a PKR 50M property deal exposes the firm to professional liability claims that dwarf any technology investment. Human reviewers working under time pressure miss an estimated **10–15% of material clauses** (ABA Legal Technology Survey, 2024). Each missed clause is a potential malpractice claim.

1.4.4 Competitive Displacement

Larger firms with greater associate bandwidth are absorbing mandates from mid-size firms that cannot match 48-hour turnaround SLAs. Firms that do not modernise in the next 2–3 years risk **permanent market share loss** as AI-assisted firms become the institutional client's default choice.

1.4.5 Client Attrition Cost

Acquiring a new corporate client costs 5–7 times more than retaining one. A firm losing 3 clients per year to competitors due to slow turnaround, each worth PKR 2M in annual fees, is losing **PKR 6M per year in client lifetime value**—not only in missed fees, but in referrals and repeat mandates that compound over years.

Total Estimated Annual Cost of Inaction per Firm	
Cost Category	Estimated Annual Impact
Lost mandates (2/month × PKR 1M avg)	PKR 24,000,000
Wasted senior associate time	PKR 13,000,000
Client attrition (3 clients × PKR 2M LTV)	PKR 6,000,000
Liability exposure (risk-adjusted)	PKR 5,000,000
Total	PKR 48,000,000

2 Proposed Solution & Technical Architecture

2.1 Solution Overview

The solution is a Retrieval-Augmented Generation (RAG) document review system specifically configured for Pakistani legal due diligence workflows. The system allows a law firm to upload a bundle of legal documents in PDF format through a simple web interface. The system indexes these documents using vector embeddings and cross-references them against a persistent knowledge base of Pakistani legal statutes — automatically generating a structured review memorandum highlighting clause-by-clause findings, identified red flags, and missing documents.

Unlike generic document Q&A tools, the system maintains a persistent knowledge base of Pakistani legal statutes including the Constitution of Pakistan 1973, Transfer of Property Act 1882, Registration Act 1908, Stamp Act 1899, Companies Act 2017, and LDA/CDA bye-laws. Every query runs against both this statutory corpus and the uploaded client documents simultaneously, enabling the system to cross-reference findings against specific legislative requirements and produce citations that include both the relevant clause in the client document and the governing Pakistani statute. Lawyers may also submit free-form queries beyond the standard checklist, allowing the system to function as an interactive legal research assistant rather than a static review tool.

Key Value Proposition

The lawyer’s role shifts from spending **5–7 days** writing a review from scratch to spending **2 hours** reviewing, correcting, and finalising an AI-generated first draft. Standard transactions achieve **same-day delivery**. Unlike generic document Q&A tools, the system knows Pakistani law — querying both the uploaded document bundle and a persistent statutory corpus simultaneously, citing specific provisions of the Constitution, Transfer of Property Act, and LDA/CDA bye-laws in every finding.

2.2 System Architecture

The pipeline consists of nine sequential layers, each with a distinct responsibility.

2.3 Dual-Index Architecture — Core Differentiator

The system maintains two separate ChromaDB indexes that are queried simultaneously on every request via LlamaIndex’s RouterQueryEngine.

Table 1: Dual-Index Architecture

Index	Contents	Lifecycle
Persistent legal corpus	Constitution of Pakistan 1973 (Articles 23, 24, 25, 172), Transfer of Property Act 1882, Registration Act 1908, Stamp Act 1899, Companies Act 2017, Muslim Family Laws Ordinance 1961, Succession Act 1925, West Pakistan Muslim Personal Law (Shariat) Application Act 1962, Benami Transactions (Prohibition) Act 2017, Anti-Money Laundering Act 2010, Income Tax Ordinance 2001 (Sections 236C & 236K), LDA/CDA/RDA/TMA/SBCA bye-laws, SBP prudential regulations, Pakistani document taxonomy (Fard-e-Malkiat, Intiqal, Registry, Aks Shajra, Jamabandi), city-specific regulatory profiles (Islamabad, Rawalpindi, Lahore, Karachi)	Built once, permanent
Session document index	Lawyer’s uploaded PDF bundle for the current transaction	Per upload, 24-hour TTL

This means every finding cites two sources: the relevant clause in the uploaded client document **and** the governing Pakistani statute. For example, a missing LDA NOC is not merely flagged as a missing document — the system identifies the specific LDA bye-law provision being violated and includes the statutory citation in the generated memo.

2.4 Pakistan-Specific Intelligence Layer

Beyond dual-index retrieval, the system incorporates five layers of Pakistan-specific legal intelligence absent from international legal AI tools such as Google NotebookLM or Kira Systems:

1. **Multilingual document handling:** The embedding model (paraphrase-multilingual-MiniLM-L12-v2) natively supports mixed Urdu-English documents — the dominant format in Pakistani legal practice. Urdu clauses, property descriptions, and Khasra references are embedded and retrieved with full semantic accuracy.
2. **Pakistani document taxonomy:** The system recognises and validates Pakistan-specific document types including Fard-e-Malkiat, Intiqal, Registry, Aks Shajra, and Jamabandi — automatically identifying which are present in an upload and which are missing from the bundle.
3. **City-specific regulatory profiles:** NOC requirements and governing authorities differ by city. The system maintains separate regulatory profiles for Islamabad (CDA), Rawalpindi (RDA/TMA), Lahore (LDA), and Karachi (SBCA), loading the correct statutory references and bye-laws automatically based on the transaction’s declared location.
4. **Housing society compliance:** For transactions involving DHA, Bahria Town, or other private housing societies, the system loads society-specific transfer documentation requirements and validates compliance accordingly — covering allocation letters, NOCs, dues clearance certificates, and membership transfer conditions.
5. **FBR withholding tax validation:** The system checks compliance with Sections 236C and 236K of the Income Tax Ordinance 2001, verifying tax challan presence, buyer and seller filer status, and the applicable withholding rate (1% for filers, 2% for non-filers) for all transactions above PKR 5,000,000.
6. **Constitutional article mapping:** Every finding is mapped to its constitutional basis under the 1973 Constitution. A missing NOC cites Article 24 (protection of property rights); a co-owner consent gap cites Article 25 (equality of citizens); an ownerless property indicator cites Article 172 (property vesting in the state). No international legal AI tool reasons at the constitutional level.
7. **Muslim inheritance and succession compliance:** The system detects inherited property indicators in uploaded documents (keywords: “late”, “deceased”, “legal heirs of”, “virasat”, “succession”) and automatically triggers a supplementary inheritance checklist validating succession certificates, legal heirship certificates, and consent of all heirs under the Muslim Family Laws Ordinance 1961 and the West Pakistan Muslim Personal Law (Shariat) Ap-

plication Act 1962. Share calculations follow Hanafi fiqh as recognised under Pakistani law.

8. **Benami transaction detection:** The system flags potential benami indicators under the Benami Transactions (Prohibition) Act 2017 — including consideration paid by a third party, price significantly below market value, and undisclosed relationships between buyer and financier. A dedicated Benami Risk Assessment section is generated in the output memo for all flagged transactions.
9. **FATF/AML compliance checks:** For transactions above PKR 10,000,000, the system triggers enhanced due diligence requirements under the Anti-Money Laundering Act 2010 and SECP AML/CFT Regulations 2018, verifying source of funds documentation and flagging Suspicious Transaction Report (STR) obligations where applicable.

2.4.1 Layer 1 — User Interface (React + Vite)

The lawyer uploads a PDF bundle via drag-and-drop and selects transaction type (property / loan / acquisition). The interface offers three operating modes via a tab selector:

- **Auto Checklist:** Runs all 15 standard due diligence questions automatically and generates the full structured memo. Suitable for standard transactions.
- **Free-Form Q&A:** The lawyer types any legal question in natural language. The dual-index engine retrieves context from both the uploaded documents and the Pakistani statutory corpus and responds with a cited finding.
- **Hybrid:** Auto-checklist runs first to generate the base memo, after which a Q&A panel opens for follow-up queries on specific findings.

A live progress bar monitors each pipeline stage. On completion, a download button delivers the memo in both Word and PDF formats.

2.4.2 Layer 2 — Backend Orchestration (FastAPI)

FastAPI receives uploaded files, assigns a UUID session identifier, manages pipeline state, and orchestrates all downstream calls. Its async-native architecture ensures concurrent uploads do not block each other. Auto-generated OpenAPI documentation at `/docs` simplifies endpoint testing during development.

2.4.3 Layer 3 — Document Processing (pdfplumber + pytesseract)

`pdfplumber` handles text-based PDFs. If text extraction returns empty or sparse content (scanned documents), the system falls back automatically to `pdf2image + pytesseract` OCR. Output is clean UTF-8 text per page with page-number metadata preserved for clause referencing in the final memo.

2.4.4 Layer 4 — Text Chunking (LlamaIndex SentenceSplitter)

`LlamaIndex SentenceSplitter` with `chunk_size=512` and `chunk_overlap=50` tokens. Metadata per chunk includes source filename, page number, and chunk index. The

50-token overlap ensures legal clause boundaries are not split across chunks, preserving context for accurate retrieval.

2.4.5 Layer 5 — Embedding Generation (HuggingFace)

`sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2` runs locally, producing 384-dimensional dense vectors with native support for mixed Urdu-English text — the dominant format in Pakistani legal documents. No API cost, no rate limits, no data leaving the machine. A 50-page document bundle embeds in under 10 seconds on CPU.

2.4.6 Layer 6 — Vector Storage (ChromaDB)

ChromaDB with a persistent local client. One collection per upload session, named with the session UUID for isolation. Native LlamaIndex `ChromaVectorStore` integration requires zero adapter code. Production upgrade path: Qdrant.

2.4.7 Layer 7 — Query Engine (LlamaIndex RouterQueryEngine)

`RouterQueryEngine` wraps two `VectorIndexRetriever` instances (persistent legal corpus + session document index), each with `similarity_top_k=5`. For every query — whether from the automated checklist or a free-form lawyer question — the engine retrieves the 5 most relevant chunks from each index and merges the context before passing it to the LLM. This dual-retrieval ensures every LLM response is grounded in both the uploaded document and the applicable Pakistani statute.

2.4.8 Layer 8 — LLM Generation (Claude Sonnet 4.6 / Groq)

Retrieved chunks, the checklist question, and a structured system prompt are sent to Claude Sonnet 4.6 (production) or Groq Llama 3.3 70B (development/testing). The LLM responds in a structured JSON format: `finding`, `clause_reference`, `page_number`, `risk_level`, and `recommendation`. Red flag detection is integrated into the same pass.

2.4.9 Layer 9 — Output Generation (python-docx + ReportLab)

`python-docx` assembles the Word memo. ReportLab generates a PDF version. The firm's letterhead is injected at the top. Memo structure: cover page → transaction summary → clause-by-clause findings → red flags section → missing documents checklist → recommendations.

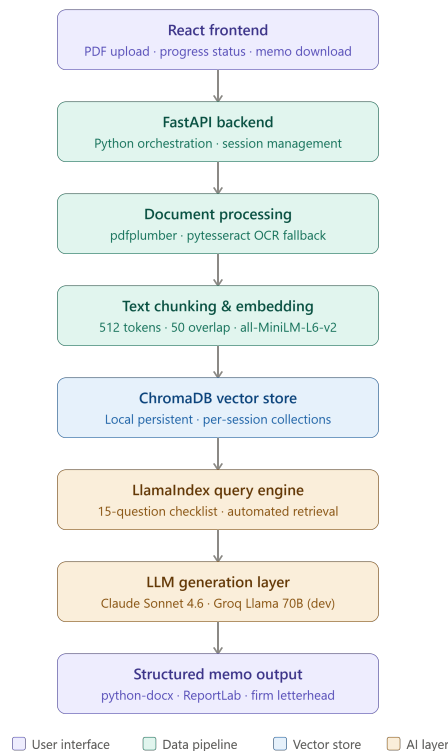


Figure 1: RAG Legal Document Review System — Complete Data Flow

2.5 Technology Stack & Selection Rationale

2.5.1 LlamaIndex (over LangChain)

LlamaIndex is selected as the orchestration framework over LangChain for the following reasons:

- **Purpose-built for document retrieval:** LlamaIndex achieved a 35% boost in retrieval accuracy in 2025 specifically for document-heavy applications—exactly this use case. LangChain is optimised for multi-step agentic workflows with tools and memory, which this pipeline does not require.
- **Lower development overhead:** Built-in query engines, routers, and fusers designed for RAG mean significantly less boilerplate than an equivalent LangChain build.
- **Gentler learning curve:** Its high-level API and focus on data connection and querying makes it faster to reach a working pipeline state—critical on a 5-day sprint.
- **Native PDF loaders via LlamaHub:** Pre-built data connectors for PDFs, databases, and APIs reduce integration work on Day 2.
- **Retrieval performance:** Benchmarks report 2–5× faster lookup times versus generic search pipelines, and 92% accuracy in retrieval benchmarks for document-heavy workflows.

2.5.2 Claude Sonnet 4.6 (Production LLM)

- **Pricing:** \$3.00 input / \$15.00 output per million tokens, translating to approximately PKR 45 per complete document review.
- **Context window:** 1 million tokens at standard pricing with no long-context surcharge—an entire document bundle fits within one context.
- **Structured output compliance:** Claude’s instruction-following is industry-leading for JSON-format generation, which the memo pipeline depends on entirely.
- **Legal document comprehension:** Consistently outperforms peers on long-document analysis, clause identification, and contradiction detection.

2.5.3 Groq Llama 3.3 70B (Development & Testing)

- **Cost:** Completely free on the developer tier—no credit card required, no per-token charge.
- **Free-tier rate limits:** 30 RPM, 1,000 RPD, 12,000 TPM, and 100,000 TPD—sufficient for all development and demo testing workflows.
- **Speed:** LPU hardware delivers 300–400 tokens per second, making demo runs impressively fast for client presentations.
- **API compatibility:** OpenAI-compatible endpoint—switching to Claude in production requires changing only the base URL and API key. Zero other code changes.

2.5.4 HuggingFace paraphrase-multilingual-MiniLM-L12-v2 (Embeddings)

Free, open-source, runs entirely locally. 384-dimensional dense vectors trained on 50+ languages including Urdu, with no API dependency, no rate limits, and no data leaving the machine. Selected over the English-only `all-MiniLM-L6-v2` specifically to handle mixed Urdu-English Pakistani legal documents — Urdu property descriptions, Khasra references, and vernacular clauses are embedded and retrieved with full semantic accuracy.

2.5.5 ChromaDB (Vector Store)

Zero-infrastructure local vector store. Installed with a single pip command and operational in three lines of Python. Native LlamaIndex `ChromaVectorStore` integration. Persistent local client handles session isolation via UUID-named collections. The production upgrade path is Qdrant, which provides a near-identical LlamaIndex API with managed cloud support and horizontal scaling.

2.5.6 FastAPI (Backend)

Async-native Python framework. Concurrent uploads handled without blocking. Auto-generated OpenAPI documentation at `/docs` for rapid endpoint testing throughout the sprint. Pydantic request validation built in with no extra configuration.

3 Due Diligence Checklist — Property Transaction

The following 15-question checklist is the automated review specification for property transaction due diligence. It is hardcoded as the initial checklist in the Day 2 build and will be extended to cover loan and acquisition transaction types on Day 3.

3.1 Property Transaction Checklist

- Q1.** Is the title deed present and registered in the name of the vendor?
- Q2.** Is the property free from all encumbrances, mortgages, charges, or liens?
- Q3.** Are all required NOCs present—LDA/CDA/TMA and all relevant local authority approvals?
- Q4.** Are the property boundaries and area measurements consistent across all submitted documents?
- Q5.** Is there any ongoing or threatened litigation, court order, or injunction against the property?
- Q6.** Is the sale consideration consistent across all documents and supported by bank evidence or payment receipts?
- Q7.** Are all signatures, attestations, and notarial certifications present and valid?
- Q8.** Is the land use classification consistent with the intended purpose of acquisition?
- Q9.** Are utility connections (electricity, gas, water, PTCL) documented and registered in the vendor's name?
- Q10.** Are all outstanding property taxes, withholding taxes, and CVT arrears disclosed and settled?
- Q11.** Is the vendor's CNIC verified and consistent across all transaction documents?
- Q12.** Are there any co-owners, inherited co-sharers, or third-party interests in the property?
- Q13.** Is the possession transfer date and handover mechanism explicitly documented?
- Q14.** Are all conditions precedent to the transfer clearly identified and satisfied?
- Q15.** Is there a registered mutation (Intiqal) on record, or is the transfer mode otherwise identified and legally valid?

3.2 Red Flag Detection Patterns

Any finding with `risk_level: HIGH` is automatically surfaced in the dedicated Red Flags section of the generated memo.

Table 2: Red Flag Detection Rules — Initial Set

Pattern	Trigger Condition	Risk Level
Missing NOC	No NOC document present or issuing authority not identified	HIGH
Price discrepancy	Sale price differs across any two documents by more than 5%	HIGH
Undisclosed litigation	Court order, injunction, or caveat found in any document	HIGH
Third-party interest	Co-owner or heir mentioned without written consent to transfer	HIGH
CNIC mismatch	Vendor CNIC number differs across any two documents	HIGH
Missing mutation	No Intiqal or equivalent transfer record present	MEDIUM

3.3 LLM Response Schema

Each checklist question produces one structured JSON object from the LLM. This schema drives both the tabular findings section and the red flag detection logic in the output memo.

```

1 {
2   "question_id": 1,
3   "question": "Is the title deed present and registered in the
4     vendor's name?",
5   "finding": "Title deed (Registry No. 4521/2023) is present
6     and registered in the name of Muhammad Asif Khan, matching
7     vendor CNIC.",
8   "reasoning": "Document 1 page 3 contains a registered sale
9     deed bearing Registry No. 4521/2023. The vendor name
10    matches the CNIC provided on page 1. Under Section 17 of
11    the Registration Act 1908, immovable property transactions
12    above PKR 100 must be registered. This requirement is
13    constitutionally grounded in Article 23, which guarantees
    the right to acquire and dispose of property subject to law
    .",
6   "document_citation": "Uploaded document 1, Page 3, Section A"
7   ,
7   "statutory_citation": "Registration Act 1908, Section 17(1)(b
8     )",
8   "constitutional_basis": "Article 23 - Right to acquire and
9     dispose of property",
9   "page_number": 3,
10  "risk_level": "LOW",          # LOW | MEDIUM | HIGH
11  "recommendation": "No action required. Verify stamp duty
12    payment receipt.",
12  "missing_documents": [],
13  "query_source": "checklist"  # checklist | freeform

```

14

```
}

```

Listing 1: Per-Question LLM Response Format

4 Cost Analysis

4.1 Per-Review API Cost Model

A standard property due diligence review involves a 50-page document bundle. The breakdown below uses Claude Sonnet 4.6 as the production LLM at current June 2026 pricing.

Table 3: Per-Review API Cost Breakdown — Claude Sonnet 4.6

Component	Tokens	Rate (USD/MTok)	Cost (USD)
Document bundle (50 pages × 500 tok)	25,000	\$3.00	\$0.075
15 checklist queries (× avg 500 tok)	7,500	\$3.00	\$0.023
System prompt overhead	1,000	\$3.00	\$0.003
LLM output (15 findings + memo draft)	4,000	\$15.00	\$0.060
Total per review	37,500	—	\$0.161
Total per review (PKR)			≈ PKR 45
Embeddings (HuggingFace, runs locally)	—	FREE	\$0.000
Vector store (ChromaDB, runs locally)	—	FREE	\$0.000

Development Cost During This Sprint

All 5-day sprint development and testing uses Groq Llama 3.3 70B on the free tier. **Total API development cost: PKR 0.** Free-tier limits (30 RPM, 12K TPM, 100K tokens/day) are sufficient for all testing and demo workflows.

4.2 Monthly SaaS Cost Projections

Table 4: Monthly Operational Cost vs Revenue (Production)

Item	Starter	Professional	Enterprise
API cost @ PKR 45/review	PKR 1,125 (25 rev.)	PKR 4,500 (100 rev.)	PKR 9,000 (est.)
Hosting (Railway paid tier)	PKR 1,400	PKR 1,400	PKR 2,800
Total COGS	PKR 2,525	PKR 5,900	PKR 11,800
Revenue	PKR 15,000	PKR 35,000	PKR 70,000
Gross Margin	83%	83%	83%

4.3 Pricing Model

Table 5: SaaS Pricing Tiers for Pakistani Law Firms

Feature	Starter — PKR 15,000/mo	Professional — PKR 35,000/mo	Enterprise — PKR 70,000/mo
Monthly reviews	25	100	Unlimited
Transaction types	Property	All 3 types	All 3 types
Custom checklist	×	✓	✓
Custom letterhead	×	×	✓
API access	×	×	✓
SLA guarantee	×	×	✓

4.3.1 Client ROI at Professional Tier

At PKR 35,000 per month, a firm processing 50 reviews per month replaces approximately PKR 500,000 per month in senior associate time. This delivers a **14× return in month one**. Even at 10 reviews per month, the subscription pays for itself within the first week of use via time savings alone.

5 Target Client Identification

5.1 Primary Targets for Day 4 Demo

5.1.1 Target 1 — Rana Ijaz & Partners (Islamabad)

- **Founded:** 1967 (formerly Rana Law Chambers)
- **Location:** Islamabad

- **Practice areas:** Legal due diligence, contracts, company incorporation, commercial litigation, domestic and international arbitration
- **Why a fit:** “Legal due diligence” is explicitly listed as a core practice area—a direct product-market match
- **Transaction type for demo:** Property due diligence
- **Outreach:** LinkedIn search “Rana Ijaz Partners Islamabad” and website contact form; request the Managing Partner or Senior Associate handling due diligence matters

5.1.2 Target 2 — M.A. Awan & Co. / MAACO (Rawalpindi & Islamabad)

- **Location:** Headquartered in Rawalpindi and Islamabad
- **Practice areas:** Corporate, commercial, real estate, banking, tax, civil, and criminal law
- **Why a fit:** Covers all three target transaction types; based in Rawalpindi making an in-person demo logistically simple
- **Transaction types for demo:** Property due diligence (primary), loan documentation review (secondary)
- **Outreach:** Lawzana.com listing for direct contact details; call and ask for the partner handling property or banking transactions

5.2 Backup Target

Mumtaz & Brohi (Islamabad) — Premier firm with a team of foreign-qualified lawyers handling transactional, advisory, and adversarial legal services for domestic and foreign clients.

5.3 Approved Outreach Script

“Salam, I am an AI engineering intern working on a legal technology project. We have built a system that automates first-pass property due diligence review—it takes 50 pages of legal documents and produces a structured review memorandum in under 5 minutes. We are looking for a law firm to conduct a live demonstration with this week. Would you have 30 minutes for a quick demonstration? There is no cost and no commitment involved.”

5.4 Client Feedback Form (Post-Demo)

Table 6: Day 4 Client Feedback Form Template

Field	Response
Firm name	
Contact person & designation	
Demo date & format	
Transaction type demonstrated	
What they liked	
Requested changes	
Interest level (1–5)	
Follow-up scheduled?	